



FINAL EXAMINATION
SEMESTER Fall 2004
Subject code – MTH 302 (Fall-2004)

Total Marks: 50
Duration:150mins

StudentID/LoginID	
Name	
PVC Name/Code	
Date	

Maximum Time Allowed: (2.5 Hours)

Please read the following instructions carefully before attempting any of the questions:

- 1.** Attempt all questions. Marks are written adjacent to each question.
- 2.** Do not ask any questions about the contents of this examination from anyone.
 - a.** If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.
 - b.** If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.
 - c.** Write all steps, missing steps may lead to deduction of marks.
 - d.** You have to solve the questions on Excel sheet and don't use more than two Excel sheets to solve all the questions.

****WARNING: Please note that Virtual University takes serious note of unfair means.**

Anyone found involved in cheating will get an `F` grade in this course.

For Teacher's use only

Question	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Total
Marks										

Question No: 1

Marks: 5

Class marks for grouped data as well as frequency against each class interval are given in the following table. Calculate using EXCEL total marks and mean.

Marks		Frequency(f)
30	34	2
35	39	6
40	44	7
45	49	16
50	54	20
55	59	8
60	64	3

Question No: 2

Marks: 5

For X and Y data calculate the StDev of X and Y, Cov(X,Y) and correlation between X and Y. Use only basic formulas in the Worksheet.

X	Y
6	120
12	200
8	150
14	170
7	150

Question No: 3

Marks: 5

Annual discount rate is 12%. Initial cost of investment 1 year from today is Rs.15000. Return from first year is Rs. 4000. Return from second year is 4500.

Return from third year is Rs. 6500. What is the net present value of this investment? Use Excel function NPV.

Question No: 4

Marks: 5

Plot the above data as actual data as well as moving averages.

Known y	Known x
2	6
3	5
9	11
1	7
8	5
7	4
5	4

Question No: 5

Marks: 10

For the data shown below find simple averages, moving averages, actual – trend and seasonal variations.

Day	Period	Data
1	Morning	270
	Afternoon	340
	Evening	330
2	Morning	276
	Afternoon	252
	Evening	333
3	Morning	282
	Afternoon	261
	Evening	342

Question No: 6

Marks:5

For the data given below find seasonal variations and plot the data as a line graph.

Qtr	Actual
1	242
2	154
3	262
4	306
1	230
2	150
3	274
4	298
1	226
2	142
3	262
4	286

Question No: 7

Marks:5

(a) There are 120 objects. The number of objects in each permutation is 4. Find the total number of permutations.

(b) Probability of success is 65%. What is the probability of 8 successes out of 9? Use BINOMDIST for the

calculation.

Question No: 8

Marks:5

- (a) Number of events are 4. Expected mean is 10. What is cumulative Poisson probability for these events?
- (b) Number of events is 4. Cumulative Poisson probability for these events is 0.19%. What is the Expected mean? Use POISSON function using hit and trial method.

Question No: 9

Marks:5

- (a) Find proportion of bags which have weight in excess of 615 g.
Mean = 610. StDev = 3.0 g.
- (b) What percentage of bags filled by the machine will weigh less than 607.5 g?
Mean = 610 g; StDev = 3.0 g.



FINAL EXAMINATION
SEMESTER Fall 2005
MTH 302 (Business Mathematics & Statistics)

Total Marks: 50
Duration:150mins

StudentID/LoginID	Bc050400073
Name	Amena Azhar
PVC Name/Code	OPKST
Date	14th March, 2006

Maximum Time Allowed: (2:30 Hours)

Please read the following instructions carefully before attempting any of the questions:

1. Attempt all questions. Marks are written adjacent to each question.
2. Do not ask any questions about the contents of this examination from anyone.
 - a. If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.
 - b. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.
 - c. Write all steps, missing steps may lead to deduction of marks.
 - d. **You have to solve the questions on Excel sheet and don't use more than two Excel sheets to solve all the questions.**
 - e. Solve the questions on excel sheet in usual manner i.e. from upward to downward not from left to right.

****WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will get an `F` grade in this course.**

For Teacher's use only						
Question	Q1	Q2	Q3	Q4	Q5	Total
Marks						

[Answers of Excel Sheet.](#)

Question No: 1

Marks: 10

Find the **product** of matrices A and B, **inverse** of A and B. Also find the **determinant** of A and B. Using EXCEL function

Where

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 4 & 5 & -6 \\ 7 & 8 & 9 \end{pmatrix}, B = \begin{pmatrix} 2 & 3 & 1 \\ 3 & 2 & 4 \\ 5 & 4 & 2 \end{pmatrix}$$

Question No: 2

Marks: 10

a). Use Excel function **NPV** to find the net present value of investment when annual discount is 9%, initial cost of investment one year from today is 150000. Return from first year is Rs 22000. Return from Second year is 45000; return from third year is 35000.

Marks 5

b). Find the **single discount rate** that is equivalent to the series 20%, 15%, 9%.

Marks 5

Question No: 3

Marks: 10

Find the **mean** and **harmonic mean** for the following data

Classes	Frequency.
30-39	2
40-49	3
50-59	11
60-69	20
70-79	32
80-89	25
90-99	7

Question No: 4

Marks: 10

For the following data :

X	3	4	5	6	7	8	9	10
Y	25	24	20	10	19	17	16	13

a) Find the regression equation by calculating **Slope** and **intercept** Excel functions.

Marks 5

b) Find **Correlation coefficient and covariance**.

Marks 5

Question No: 5

Marks: 10

a) For the following data

Marks 5

Day	Period.	Data
1	Morning	68
	Afternoon	29
	Evening	78
2	Morning	41
	Afternoon	72
	Evening	61
3	Morning	58
	Afternoon	89
	Evening	60

a). Find the **moving average**. Also plot **line graph** of moving averages.

Marks 5

b). A Surgery is successful for 55% patients. What is the probability of its success in at least 3 cases out of randomly selected 5 patients. Use **BINOMDIST** of Excel function.

Marks 5

	FINAL TERM EXAMINATION FALL 2007 MTH302 - BUSINESS MATHEMATICS & STATISTICS (Session - 3)	Marks: 60
StudentID/LoginID: _____ Student Name: _____ Center Name/Code: _____ Exam Date: Saturday, March 22, 2008		
Please read the following instructions carefully before attempting the paper: Duration : March 22, 2008 to March 24, 2008. 1- Take-home exams allow you to use notes, texts or resource materials while solving the paper. Take-home exams test your ability to find and use information for problem solving and to deliver well-structured and well-presented arguments and solutions. <u>All students must submit their paper on or before 24th March 2008. No relaxation in the time for the submission shall be given</u> 3- Attempt all the questions on one Excel Worksheet (use only sheet 1) to solve all the questions. More than one sheet is not acceptable. If you solve any question on sheet2, sheet3 and so on, it will not be graded. 4- Solve the questions on excel worksheet in usual manner i.e. from upward to downward not from left to right. 5- Variables (say x, y, and z) are given in the questions you can assign any value to variable within the range mentioned in the questions. (e.g. A man sells shirts of Rs x and he gets a profit of 10%. Then find the profit. ($25000 < x < 30000$)) Solution: Here we can take x as any value between 25000 to 30000. Take x = 26000 Profit = $26000 \times 10 / 100 = 2600$ Rs Answer Or Take x = 26500 Profit = $26500 \times 10 / 100 = 2650$ Rs Answer)		

6- Attempt all questions. Marks are written adjacent to each question.

7- Do not ask any questions about the contents of this examination from anyone.

If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.

If you think that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.

8- Write all steps, missing steps may lead to deduction of marks.

9- you do not solve paper on Excel Sheet, you will get a zero mark. So paper on Word files or any format other than Excel will not be graded.

For Teacher's use only										
Question Marks	1	2	3	4	5	6				Total

Question No: 1 (Marks: 12)

- a)** A shopkeeper borrowed Rs 20000 from two money lenders, he paid $X\%$ interest on first loan and $(X+2)\%$ interest on second loan per annum. Where $6 \leq X \leq 25$. After one year if he paid rupees 2500 in total then how much rupees he borrowed from each lender.

Marks 4

- b)** Find the compounded interest rate percent per annum, if Rs 2000 amounted to Rs 3000 in y years ; $y = 2, 3, 4$ (choose any one of them), Interest is considered half –yearly.

Marks 4

- c)** Take x number of persons in any three families and divide each family in three categories **No of men, No of women, No of children**, and make their matrix. (where $6 \leq X \leq 17$)

Since the recommended daily requirement of calories for three categories is

Men = 2500gm, Women = 2000gm, Children = 1800gm,

And the requirement of protein for three categories is

Men = 55gm, Women = 45gm, children = 35gm,

Find the total requirement of calories and protein for each family, by using multiplication function of matrices in Excel.

Marks 4

Question No: 2 (Marks: 8)

- a)** Suppose you purchase a car, the cost of car is last six digits of your VU ID, if useful life of the car is 5 years and the value of car at the end of depreciation is 25% of cost of the car then finds depreciation for each year by using Excel function.

Marks 5

b) Suppose a set of 50 cards numbered with 20 to 70 (Take integers only) , select 10 cards from them and make a data array of integers then find Mean, Median and Mode of the data.

Marks 3

Question No: 3 (Marks: 10)

Take the each batsman score of batting line up of one inning from any cricket match you like in which at least 10 batsmen had made scores, make the data array of scores (Runs), from this data,

a) Find standard deviation and Variance.

Marks

5

b) Find all Quartiles and Quartile Deviation.

Marks

5

Question No: 4 (Marks: 12)

The National Logistic Cell (NLC) is studying the relationship between the number of bidders on a highway project and the winning (lowest) bid for the project.

i) Sketch the original data on the graph.

Marks 2

ii) Determine is there any degree of association between number of bidders and the winning (lowest) bid? And what type of association is there between the number of bidders and the winning (lowest) bid.

Marks 4

iii) Determine the mathematical equation of the winning (lowest) bid on the number of bidders on a highway project. Then also estimate the amount of the winning bid if there are seven bidders.

Marks 6

X=Number of Bidders	Y=Winning Bid (\$)
19	L(1000000)
19	(L+3)1000000
13	(L+4.7)1000000
20	(L+2.8)1000000
15	(L+2.7)1000000
20	(L+0.5)1000000
17	(L+3.3)1000000
21	(L+5.2)1000000
16	(L+0.5)1000000

16	(L+5.3)1000000
14	(L+3.0)1000000
17	(L+3.8)1000000
17	(L+4.4)1000000
17	(L+3.6)1000000
16	(L+3.1)1000000

Note: Where $5 < L < 10$

Question No: 5 (Marks: 10)

A multinational company produces three popular brands of detergents; Brand 1 and Brand 2 are premium while Brand 3 is economy in quality. To make effective decisions in the future company would like to test whether there is an association between the income level of the consumers and choice of brand at 5% level of significance. You are hired for this task to sample Z consumers in your town using random sampling procedure to cover the entire market using income as the basis of selection. The categories that were used in classifying income level are: Lower, Middle and High. ($400 < Z < 600$)

Income	Brands			Total
	Brand1	Brand2	Brand3	
Lower				200
Middle				
Upper				
Total	150			Z

Question No: 6 (Marks: 8)

Azam is taking a multiple choice online test in which each question has 4 choices. If Azam has no knowledge of correct of any of the questions then he decided for a strategy in which he will place 4 slips (marked a, b, c, d) in the packet. He randomly selects one slip for each question and replaces the slips in the packet. The marking (a, b, c ,d) on the slip will determine his answer. And probability for getting right answer is P . Also total questions in the exam are N . ($0.2 < P < 0.5$ and $20 < N < 55$)

What is the chance that

- (a) At least 15 questions are correct.
- (b) Exactly 10 questions are correct.
- (c) At most 6 questions are correct.
- (d) 5 questions are incorrect.

